# Lab Experiment-18

**Aim**

To write a Python program to implement a simple First-Order Predicate Calculus (FOPC) parser to identify and validate basic logical expressions using predicates, variables, and logical operators.

**Algorithm**

1. Get a logical expression as input from the user.
2. Define the valid logical operators such as AND, OR, NOT, and implication.
3. Identify predicates and their arguments.
4. Check whether the expression follows the basic FOPC syntax.
5. Display whether the expression is valid or invalid.
6. Print the result.

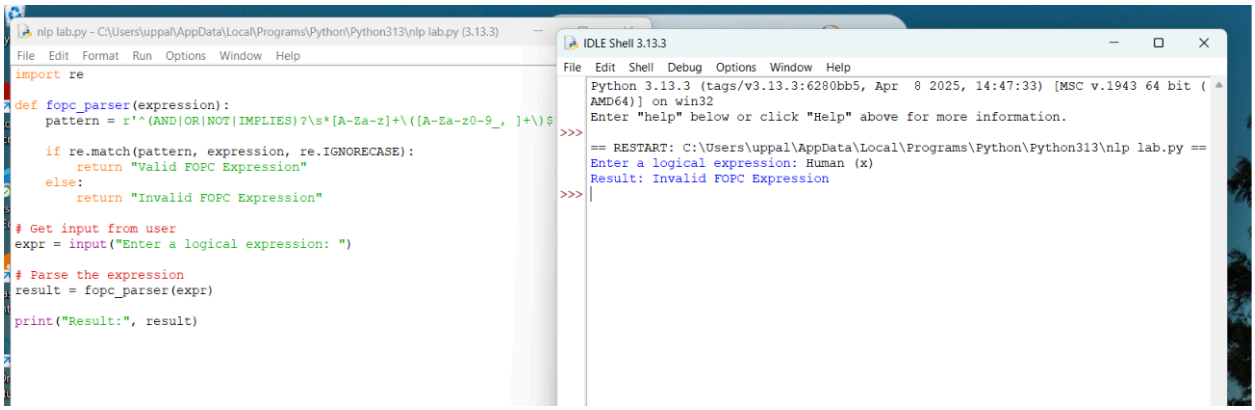
**Python Code**

```
import re

def fopc_parser(expression):
    pattern = r'^(AND|OR|NOT|IMPLIES)?\s*[A-Za-z]+[A − Za − z0 − 9] +$'
    if re.match(pattern, expression, re.IGNORECASE):
        return "Valid FOPC Expression"
    else:
        return "Invalid FOPC Expression"
expr = input("Enter a logical expression: ")
result = fopc_parser(expr)
```

print("Result:", result)

**Output:**




**Result**

Thus, the Python program successfully implements a simple FOPC parser to check basic logical expressions.